24 June 1955

SUPERSEDING
NAVCORD OS 2903A
20 October 1943

MILITARY SPECIFICATION

SIGNAL LIGHT, PISTOL CARTRIDGE, MK 2 (WHITE), MK 2 (GREEN), AND MK 2 (RED)

1. SCOPE

l.l Scope. -- This specification governs the manufacture, assembly and preparation for delivery of Pistol Cartridge Signal LIGHT Mk 2 (White), Mk 2 (Green) and Mk 2 (Red) and the methods of inspection and tests upon which acceptance will be based. The signal cartridges covered by this specification consist of a commercial primed shot gun shell No. 10 gage containing a propelling charge and a white, green, or red pyrotechnic pellet as required. The component parts are secured in the case by crimping the end of the shell body over a felt wad and cardboard retaining disc.

1.2 Classification, -- Pistol Cartridge Signal Cartridges shall be of the following types:

Type I -- Pistol Cartridge Signal Light Mk 2 (White)
Type II -- Pistol Cartridge Signal Light Mk 2 (Green)
Type III -- Pistol Cartridge Signal Light Mk 2 (Red)

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621 --- Boxes; Wood, Nailed and Lock-Corner.
TT-S-271 --- Shellac, "Dry" (Orange) and Other Lacs.

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VV-W-95 -- Wax, Paraffin, Technical.
LL-B-631 -- Boxes; Fiber Corrugated (For Domestic Shipment).
PPP-B-601 -- Boxes; Wood, Cleated-Plywood.
PPP-B-676 -- Boxes; Set-up, Paperboard.

MILITARY

JAN-P-120 -- Packaging and Packing for Overseas Shipment-Cartons, Folding, Paperboard.
JAN-P-150 -- Potassium Chlorate (For Use in Ammunition).
JAN-A-159 -- Antimony Sulfide (For Use in Ammunition).
JAN-B-162 -- Barium Nitrate.
JAN-P-223 -- Powder, Black.
JAN-D-232 -- Dextrin (For Use in Ammunition).
JAN-A-463 -- Alcohol, Ethyl (For ordnance Use).
JAN-S-487 -- Sulfur, Ground (For Use in Ammunition).

STANDARDS

MILITARY

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -- Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDANCE

LD SK 102941 -- Pistol Cartridge Signal Light Mk 2 (White) Mk 2 (Green), and Mk 2 (Red) and all documents listed thereon.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

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2.2 Other publications. -- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 -- Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing office, Washington 25, D. C. Orders for the above publication should cite 49 CFR 71-78 (Rev. 1950).")

3* REQUIREMENTS

3.1 General.

3.1.1 Pistol Cartridge Signal Light Mk 2 (White), Mk 2 (Green), and Mk 2 (Red) shall be manufactured in accordance with Bureau of Ordnance List of' Drawings LD SK No. 102941 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Materials. -- Unless otherwise specified herein, materials used in the manufacture of the signals shall conform" to the requirements of the applicable drawings.

3.2.1 Black powder. -- Black powder shall conform to the requirements of Specification JAN-P-223, Grade A-3 or Grade A-4 as required.

.3.2.2 Chemicals. -- Chemicals used in the manufacture of pyrotechnic must be free from any hydroscopic or deliquescent impurities. Any chemical may be rejected if it contains impurities which do not meet these requirements.

3.2.2.1 Potassium chlorate. -- Potassium chlorate shall be in accordance with the requi-rements of Specification JAN-P-160, Grade I, Class d.

3.2.2.2 Barium chlorate. -- Barium chlorate shall contain not more than 0.2 percent chloride figured as barium chloride nor more than 0.2 percent moisture. All shall pass through a U. S. Standard Screen Scale No. 100 sieve having uniform square openings of 0.0059 inch.

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3.2.2.3 Barium nitrate. -- Barium nitrate shall conform to the requirements of Specification JAN-B-162, Class E, except that it shall all pass through a U. S. Standard Screen Scale No. 200 sieve having uniform square openings of 0.0029 inch.

3.2.2.4 Potassium nitrate. -- Potassium nitrate shall be at least 98.5 percent potassium nitrate and contain not more than 0.5 percent moisture. All shall pass through a U. S. Standard Screen Scale No. 200 sieve having uniform square openings of 0.0029 inch.

3.2.2.5 Strontium carbonate. -- Strontium carbonate shall be of a good technical grade for use in pyrotechnic mixtures. All shall pass through a U. S. Standard Screen Scale No. 200 sieve having uniform square openings of 0.0029 inch.

3.2.2.6 Antimony sulphide, -- Antimony sulphide shall conform to the requirements of Specification JAN-A-159, Grade I, except that it shall all pass through a U. S. Standard Screen Scale No. 200 sieve having uniform square openings of 0.0029 inch.

3.2.2.7 Sulphur. -- Sulphur shall conform to the requirement of Specification JAN-S-487, Grade D.

3.2.2.8 Shellac. -- Shellac shall conform to the requirements of Specification TT-S-271, Type I, Form a, Grade D, ground so that all shall pass through a U. S. Standard Screen Scale No. 20 sieve having uniform square openings of 0.0331 inch.

3.2.2.9 Alcohol. -- Alcohol shall conform to the requirements of Specification JAN-A-463.

3.2.2.10 Paraffin wax. -- Paraffin wax shall conform to the requirements of Specification VV-W-95, Type 1, Grade B.

3.2.2.11 Dextrin, -- Detrin shall conform to the requirements of Specification JAN-D-232, Type 11.

3.3 Pyrotechnic mixing. -- All material for the pyrotechnic shall be first thoroughly dried and passed through a U. S. Standard Screen Scale No. 40 sieve having uniform square openings of 0.0165 inch to remove lumps.

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3.4 Functioning. -- Upon firing of the signal cartridge, vertically, the pellet shall ascend to a minimum altitude of 200 feet and have a minimum burning time of six seconds. Testing and acceptance shall be in accordance with 4.5.

3 5 Workmanship. -- Only the best trade practices shall be utilized in the manufacture of Pistol Cartridge Signal Light Mk 2 (White), Mk 2 (Green), and Mk 2 (Red). Care shall be exercises in the assembly of the component parts to assure proper seating . and alignment, especially where crimping is employed to maintain parts in their relative positions. There shall be no burrs or other imperfections of manufacture harmfully affecting the functioning or assembly of the product. All parts shall be free of chips, dirt, and other foreign materials. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent which is employed.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of' the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot. -- As applied to the Government Inspection of units of' product, the term "Lot" shall mean 'inspection Lot", i.e., a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in **Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of ten (10) Pistol Cartridge Signal Lights Mk 2 of the prescribed color shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These cartridges shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance.

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Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Production lot. -- For procurement purposes the production lot shall consist of 10,000 cartridges of one color plus those required for all test purposes. Unless otherwise stated in the contract, order, or requisition, all units for test purposes shall be provided at the expense of the Contractor. The production lot should not be confused with the lot for inspection purposes as defined in 4..1.1.

4.4 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment, OF 400; Government Inspection shall be in accordance with the provisions of General Procedures for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OF 400.

4.5 Functioning test. -- Select and test samples from each production lot in accordance with Multiple Sampling Plan, AQL 6.5 percent defective, Code Letter G of Standard MIL-STD-105. The test shall consist of firing the signal cartridge vertically from a standard signal pistol.

4.5.1 Acceptance criteria. -- Failures shall be defined as. follows:

(a) Failure of the primer to fire.
(b) Failure of the pyrotechnic to ignite and attain a minimum altitude of 200 feet.
(C) Failure of the pellet to burn for at least six seconds with a bright light.
(d) Pellet explodes or breaks UP in flight.
(e) Pellet goes out before reaching the top of its trajectory or before the composition has been consumed.
(f) Pellet fails to light up within 200 feet of the pistol.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. -- Ten (10) Pistol Cartridge Signal Lights shall be packaged in a folding carton or set-up box, with separators, conforming to the requirements of Specifications JAN-P-120 or PPP-B-676.

5.1.1 Twenty-five (25) packages, as specified in 5.1, shall be intermediately packaged in a style RSC fiberboard box conforming to the requirements of Specification LLL-B-631.

5.2 Packing. -- Four (4) intermediate boxes, as specified in 5.1.1, shall be packed in a wood cleated plywood or nailed wood box conforming to the requirements of Specifications PPP-B-601 or NN-B-621, respectively.

5.3 Marking. -- In addition to any special marking required by the contract or order each carton, carton containers and wooden packing box shall be marked as indicated on the pertinent drawings and shall meet the requirements of' Standard MIL-STD-129 and Code of Federal Regulations 4..9 CFR 71-78.

6. NOTES

6.1 Ordering data. -- Procurement documents should specify:

(a) Title, symbol, and date of this specification.
(b) Type of Pistol Cartridge Signal Light Mk 2 required (see 1.2).

Patent notice, -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture. use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obining information on the use of this specification which will insure that suitable products can be procured with a nimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on nes on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

ECIFICATION

GANIZATION *(Of submitter)* | CITY AND STATE

NTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

TERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

JBMITTED BY *(Printed or typed name and activity)* | DATE

FORM 1126 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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